

DESIGN BASIS REPORT

PROJECT: KLIPFONTEIN 132/33 KV SUBSTATION

CLIENT: EKURHULENI METROPOLITAN MUNICIPALITY (DISTRIBUTOR)
IN CONJUNCTION WITH ESKOM TRANSMISSION

PREPARED FOR: SENIOR REVIEW / TENDER SUBMISSION / INTERNAL APPROVAL

PREPARED BY

H. MANGENA

hillarymangena@ymail.com
+27678766636

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Prepared by:

H. Mangena

Substation Engineer

Reviewed by:

DESIGN BASIS REPORT (DBR)
Klipfontein 132/33 kV Substation

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1. INTRODUCTION

This Design Basis Report (DBR) has been prepared for the proposed Klipfontein 132/33 kV Substation, a greenfield bulk supply point located on Farm Klipfontein 24 IR, near Benoni, Gauteng Province (coordinates 26°11'24" S, 28°18'36" E). The substation forms part of Eskom Transmission's Gauteng East reinforcement programme and will interface directly with Ekurhuleni Metropolitan Municipality's distribution network. Its primary purpose is to accommodate forecast load growth in the residential and light-industrial corridors east of Johannesburg, while simultaneously offloading the heavily loaded Rynfield and Daveyton 132/33 kV substations. The ultimate firm capacity target is 135 MVA at the 33 kV bus, with full N-1 security of supply in accordance with the South African Grid Code. The report is developed on the basis of the complete "Data Provided" package issued by the Client and Eskom, which comprises the following input documents (all dated and referenced in the project data room):

- 01_Network_Planning_Report.pdf
- 02_Equipment_&_Primary_Plant_Specification.pdf
- 03_Geotechnical_Report.pdf
- 04_Environmental_Impact_Assessment_Report.pdf
- 05_System_Studies_Report.pdf
- 06_Line_Design_Report.pdf
- 07_Grid_Connection_Requirements.pdf
- 08_Project_Budget_&_Scope.pdf
- 09_Protection_Philosophy_Document.pdf

This DBR establishes the fundamental design parameters, engineering philosophies, and key technical decisions that will govern all subsequent detailed engineering, procurement, construction, and commissioning activities. It translates the Client's requirements, network studies, site constraints, and statutory obligations into a coherent, auditable basis for the project. All later deliverables — including the Single Line Diagram, General Arrangement layout, earthing design, protection coordination study, and civil/structural drawings — will be developed strictly in accordance with the philosophies and selections locked in this document. The DBR has been prepared in compliance with the latest applicable revisions of the South African Grid Code, SANS 1019, IEC 62271 series, IEEE 80, and Eskom's internal technical reference specifications.

2. OBJECTIVES & SCOPE OF WORK

The primary objective of the Klipfontein 132/33 kV Substation is to establish a new N-1 compliant bulk supply point capable of delivering a firm capacity of 135 MVA at the 33 kV bus by Year 10 (2036), in accordance with the load forecast and security requirements defined in 01_Network_Planning_Report.pdf. The substation will provide reliable 33 kV distribution to support residential and light-industrial growth in the Greater Johannesburg East region while relieving overload conditions at the existing Rynfield and Daveyton 132/33 kV substations. All design decisions in this DBR are therefore driven by the need to achieve full N-1 redundancy (no loss of supply for the outage of any single 132 kV circuit, transformer, or bus section) while maintaining compliance with the South African Grid Code and NRS 048 power quality limits.

Scope of Work – Inclusions

The design and engineering covered by this DBR include:

- Complete 132/33 kV substation layout, primary plant, protection, control, SCADA, and auxiliary systems for the initial configuration of 2 × 75 MVA transformers, 2 × 132 kV incoming lines, and 8 × 33 kV outgoing feeders.
- Provision for future expansion: space and civil interfaces reserved for a third 75 MVA transformer and four additional 33 kV feeders.
- Integration of the specified 20 MVar 132 kV shunt reactor and 10 MVar 33 kV capacitor bank for reactive power management.
- Earthing system, lightning protection, station AC/DC supplies, control building, and all associated civil/structural interfaces.
- Full compliance with the environmental constraints and mitigation measures stated in 04_Environmental_Impact_Assessment_Report.pdf.

Scope of Work – Exclusions

The following items are explicitly excluded from this DBR and will be provided under separate contracts:

- Design, supply, and construction of the two incoming 132 kV overhead lines (approximately 8 km total) – covered by 06_Line_Design_Report.pdf and a separate Eskom Transmission Lines contract.
- Land acquisition and boundary fencing (already secured by the Client).
- Fibre-optic telecommunication links beyond the substation boundary.
- Any generation connection or IPP interface works.

This DBR defines the substation and immediate interfaces, ensuring design deliverables stay within the ZAR 195 million budget (08_Project_Budget_&_Scope.pdf) while meeting Network Planning Report requirements.

3. APPLICABLE STANDARDS, CODES & REFERENCES

The design, engineering, and subsequent implementation of the Klipfontein 132/33 kV Substation shall comply with all applicable national and international standards, codes of practice, and utility-specific technical requirements. These documents establish the minimum acceptable criteria for system performance, safety, reliability, insulation coordination, equipment ratings, short-circuit withstand, earthing, protection, and operational compliance. In all cases, the latest approved revisions in force as of April 2026 shall apply, unless otherwise specified by the Client or Eskom. Where conflicts arise between standards, the more stringent requirement shall take precedence, subject to formal agreement with the Client and Eskom. The primary governing documents are summarised below:

Standard / Code	Edition / Reference	Application to this Project
South African Grid Code (Network Code)	Latest revision 2025	Defines N-1 reliability, connection conditions, fault ride-through, system security and grid interface requirements
NRS 048-2	Latest edition 2024	Voltage quality limits (steady-state voltage $\pm 10\%$, flicker $P_{\text{It}} \leq 1.0$, THD $\leq 8\%$, unbalance $\leq 2\%$) that govern reactive compensation, OLTC settings and power quality compliance
SANS 1019	Latest edition 2023	Insulation coordination, voltage levels, clearance requirements, BIL selection and surge withstand for 132 kV and 33 kV plant
IEC 60076 & NRS 054	Latest editions 2022–2025	Power transformer design, rating, testing, cooling, OLTC performance and tertiary winding specifications
IEC 62271 series & SANS 62271	Latest editions 2024	High-voltage and medium-voltage switchgear ratings, short-circuit withstand, mechanical endurance and busbar design
IEC 60909-0	Latest edition 2016 (with SA amendments)	Short-circuit current calculations used to confirm all fault levels and equipment ratings
IEEE 80 & SANS 10199	IEEE 2013 / SANS latest 2023	Substation earthing grid design, step and touch voltage limits, and application of site-specific soil resistivity
NRS 037 (Parts 1–3)	Latest edition 2022	Protection philosophy, relay coordination and auto-reclose requirements
Eskom DST_34-462 & TRS / STD Series	Latest issued versions	Eskom-specific protection, control, SCADA integration, station auxiliary supplies, materials and construction practices
IEC 61850 (all parts)	Latest edition 2023	Substation communication architecture, GOOSE messaging and IEC 61850-based protection & control

The design will be subject to independent technical audit against the above documents prior to construction release. Any proposed deviation will be explicitly declared and justified in subsequent detailed design deliverables.

4. DESIGN INPUTS & DATA SUMMARY

This section provides a consolidated summary of all technical inputs received from the Client and Eskom in the “Data Provided” package. The data summarised below form the sole basis for the design philosophies and decisions presented in the remainder of this DBR. No assumptions or reinterpretations have been introduced; every parameter is extracted directly from the referenced documents.

4.1 Network Planning Data (01_Network_Planning_Report.pdf)

Parameter	Value	Notes / Requirement
132 kV 3-phase fault level	31.5 kA (3 s)	Governs all 132 kV equipment short-circuit ratings
132 kV 1-phase fault level	25 kA (3 s)	Used for earth-fault calculations and earthing design
33 kV 3-phase fault level	20 kA (3 s)	Governs 33 kV switchgear and busbar ratings
33 kV 1-phase fault level	16 kA (3 s)	–
Load forecast – Year 1 (2027)	65 MVA	Initial firm capacity
Load forecast – Year 5 (2031)	98 MVA	–
Load forecast – Year 10 (2036)	135 MVA	Ultimate firm capacity at 33 kV bus
Reliability requirement	Full N-1 compliance (no load loss for single contingency)	South African Grid Code urban category
132 kV infeed lines	2 (from different sources)	–
33 kV outgoing feeders	8 initial (provision for 12 ultimate)	–

4.2 Primary Plant & Equipment Ratings (02_Equipment_& Primary_Plant_Specification.pdf)

Item	Specification
Power transformers	2 × 75 MVA ONAN/ONAF, 132/33 kV, YNd11, Z = 12 % ± 7.5 %, OLTC ±10 % (17 steps)
Tertiary winding	11 kV, 5 MVA, BIL 95 kV
132 kV busbar rating	3 150 A continuous, 31.5 kA / 3 s
33 kV busbar rating	2 500 A continuous, 20 kA / 3 s
132 kV shunt reactor	20 MVar (line reactor)
Neutral reactor	400 Ω
33 kV capacitor bank	10 MVar (switched)

4.3 Site & Geotechnical Data (03_Geotechnical_Report.pdf)

Parameter	Value	Application
Average soil resistivity (Wenner method)	85 $\Omega \cdot m$ (wet-season design value)	IEEE 80 / SANS 10199 earthing grid design
Soil type	Clayey sand with high moisture retention	Good for earthing, moderate bearing capacity
Safe bearing capacity	180 kPa	Foundation design
Groundwater level	2.8m below surface (wet season)	Drainage and foundation considerations

4.4 Environmental & Other Constraints

Category	Description	Reference Document
Environmental constraints	Construction hours 07:00–17:00; dust suppression required; 50 m wetland buffer on eastern boundary	04_Environmental_Impact_Assessment_Report.pdf
Incoming line conductor	ACSR Zebra	06_Line_Design_Report.pdf
Project budget	ZAR 195 million (excluding VAT)	08_Project_Budget_&Scope.pdf
Grid connection & SCADA	Compliance with grid code, SCADA integration, communication protocols	07_Grid_Connection_Requirements.pdf
Protection philosophy	System-wide protection approach including line, transformer, feeder and busbar protection	09_Protection_Philosophy_Document.pdf

All subsequent engineering (busbar scheme, bay layout, earthing grid, protection settings, etc.) is developed strictly from the above inputs.

5. DESIGN APPROACH

The overall design philosophy for the Klipfontein 132/33 kV Substation is to deliver a safe, reliable, operable, and future-proof installation that fully satisfies the N-1 security and power-quality requirements of the South African Grid Code while remaining within the approved ZAR 195 million budget envelope. All philosophies are derived directly from the Data Provided package and are structured to minimise operational risk, simplify maintenance, and allow seamless future expansion without major civil or outage works.

5.1 Reliability & Security

Full N-1 compliance is mandatory: the loss of any single 132 kV circuit, transformer, bus section, or bus coupler must not result in loss of supply to the 33 kV load. This is achieved through duplicated infeed paths, redundant transformers, and busbar arrangements that permit immediate load transfer.

5.2 Operational Philosophy

The substation will be designed for high operational flexibility. Circuit breakers and disconnectors will allow isolation and maintenance of any major item of plant without requiring a full substation outage. Hot maintenance on busbars and couplers will be facilitated where possible. All switching operations will be supported by comprehensive interlocking and synchro-check supervision.

5.3 Expandability & Maintainability

Civil foundations, cable trenches, and yard space are sized for a future third 75 MVA transformer and four additional 33 kV feeders. Modular bay design and standardised equipment layouts will ensure that future extensions can be executed with minimum disruption to existing circuits.

5.4 Safety & Environmental Considerations

Personnel safety is paramount. Earthing, lightning protection, and clearances will comply with the most stringent requirements of IEEE 80, SANS 10199, and SANS 1019. Environmental constraints from the EIA (construction hours, dust suppression, wetland buffer) are incorporated into the layout philosophy from the outset.

6. ELECTRICAL CONFIGURATION & PLANT SELECTION

6.1 Voltage Levels & System Parameters

- Primary: 132 kV (transmission infeed)
- Secondary: 33 kV (distribution)
- Auxiliary: 415/230 V AC station services, 110 V DC protection/control
- System frequency: 50 Hz
- Fault levels and load forecast as summarised in Section 4.

6.2 Busbar Scheme Selection

Four standard schemes were evaluated against the project requirements of N-1 security, operational flexibility, maintainability, cost, and Eskom precedent for 132/33 kV substations.

Scheme	Reliability (N-1)	Maintenance Flexibility	Outage Impact (Bus Fault)	Relative Cost	Eskom Precedent at this Voltage	Selected
Single Bus	Low	Poor	Full substation outage	Lowest	Not used for new 132 kV substations	No
Main & Transfer Bus	High	Good (transfer bus available)	Main bus only affected	Medium	Standard configuration for 132 kV infeed substations	Yes (132 kV)
Double Bus Single Breaker	High	Excellent	One bus only affected	Medium	Standard configuration for 33 kV distribution switchgear	Yes (33 kV)
1½ Breaker Scheme	Very High	Excellent	Minimal impact	Highest	Typically used on critical 400 kV substations only	No

Selected Schemes

- 132 kV side: Main & Transfer Bus (with dedicated bus coupler). This is the Eskom-standard arrangement for transmission infeed substations. It provides full breaker maintenance capability via the transfer bus while keeping capital cost and footprint within budget.
- 33 kV side: Double Bus Single Breaker (with bus coupler). This matches the Client's operational preference and Eskom distribution practice. It gives operators the ability to transfer any feeder between Bus I and Bus II without interruption, satisfying N-1 and allowing maintenance on either busbar.

6.3 Bay Allocation & Quantities

132 kV Switchyard (AIS, 7 bays total):

- 2 × Incoming Line Bays
- 2 × Transformer Bays
- 1 × Bus Coupler Bay
- 1 × VT / Metering Bay
- 1 × Spare / Future Bay

33 kV Switchgear (indoor, 16 bays total):

- 2 × Transformer Incomer Bays
- 1 × Bus Coupler Bay
- 8 × Outgoing Feeder Bays
- 1 × 10 MVar Capacitor Bank Bay
- 1 × VT / Metering Bay
- 2 × Spare / Future Bays

6.4 Transformer Configuration

Two 75 MVA ONAN/ONAF transformers (Section 4.2) operating in parallel with equal load sharing under normal conditions. Each transformer is sized to carry the full Year-10 firm load under N-1. OLTC control will be automatic via AVR with manual override. Tertiary 11 kV winding is provided for possible neutral reactor connection or future auxiliary supply.

6.5 Reactive Power Compensation

- One 20 MVar 132 kV shunt reactor (line reactor) on the incoming circuits to control voltage rise.
- One 10 MVar switched 33 kV capacitor bank for voltage support and power-factor correction under peak load.

Conceptual High-Level Configuration (basis for detailed SLD)

Power flows from two independent 132 kV lines onto the Main Bus → Bus Coupler → Transfer Bus → two transformer bays → 33 kV Double Bus system → eight outgoing feeders plus capacitor bank. Full switching flexibility exists between buses on both voltage levels.

7. EARTHING & LIGHTNING PROTECTION PHILOSOPHY

The substation earthing system will be designed in accordance with IEEE Std 80 (2013) and SANS 10199 using the site-specific soil resistivity of $85 \Omega \cdot \text{m}$ (Section 4.3). A single integrated earthing grid will be provided beneath the entire switchyard and control building, connecting all metallic structures, equipment frames, transformer neutrals, and surge arresters to achieve an equipotential plane.

7.1 Earthing Grid Design Criteria

- Maximum grid resistance $\leq 1.0 \Omega$.
- Step voltage $\leq 2.5 \text{ kV}$, touch voltage $\leq 0.7 \text{ kV}$ (IEEE 80 tolerable limits for 0.5 s fault clearance).
- Grid layout: $10 \text{ m} \times 10 \text{ m}$ mesh with 3 m vertical rods at perimeter and high-current injection points.
- Additional counterpoise conductors where local resistivity exceeds $100 \Omega \cdot \text{m}$.
- All connections by exothermic welding or approved mechanical clamps.
- Transformer neutral earthing: solidly earthed at 33 kV, reactor-neutral resistor at 132 kV as per protection philosophy.

7.2 Lightning Protection Philosophy

- Shield wires and/or lightning masts will be provided to achieve a shielding angle of $\leq 30^\circ$ over all 132 kV and 33 kV equipment.
- Surge arresters (metal-oxide, station class) will be installed at line entrances, transformer bushings, and capacitor bank terminals, coordinated per SANS 1019 and IEC 60099.
- Rolling-sphere method (20 m radius) will be used for final mast placement.

7.3 Safety & Interface Requirements

- The earthing grid will extend 2 m beyond the substation fence and be bonded to it. Fence earthing will be independent of the main grid to prevent transferred potentials. All cable trenches will include earthing conductors bonded at both ends. The design will be verified by detailed grid calculations and step/touch voltage contour plots prior to construction.

8. PROTECTION, CONTROL & AUTOMATION PHILOSOPHY

The protection, control and automation systems for Klipfontein 132/33 kV Substation are designed to provide fast, selective and reliable fault clearance while ensuring full compliance with the protection philosophy in 09_Protection_Philosophy_Document.pdf, NRS 037 and Eskom DST_34-462. All schemes will be implemented using numerical multi-function relays with IEC 61850 GOOSE messaging for hardwired inter-bay interlocking and trip circuits.

8.1 Protection Schemes

- 132 kV incoming lines: Main distance protection (21/21N) with permissive over-reach transfer trip (POTT) and backup directional OC/EF. Auto-reclose with synchro-check.
- Transformers: Biased differential (87T), restricted earth fault (64REF), overfluxing (24), Buchholz and sudden pressure, with OLTC supervision.
- 33 kV feeders: Directional OC/EF (67/67N) with sensitive earth fault and auto-reclose.
- Busbar: Low-impedance differential protection on both 132 kV and 33 kV buses (high-impedance backup considered for 33 kV).
- Breaker failure: Dedicated BF protection on all 132 kV and 33 kV circuit breakers with local and remote trip initiation.

8.2 Control & Automation Philosophy

- Bay-level control will be provided via bay control units (BCU) with full local HMI on each bay. Station-level control will be centralised in the control room with redundant station controllers. All control actions will be subject to comprehensive interlocking, synchro-check and pole-discrepancy supervision. Automatic busbar transfer and load-shedding logic will be implemented for N-1 contingencies.
- Protection grading and coordination will be confirmed in the separate Protection Coordination Study. All relays will be IEC 61850 compliant with full event recording, disturbance recording and time synchronisation via PTP (IEEE 1588).

9. SCADA, TELECOMMUNICATION & METERING PHILOSOPHY

SCADA and telecommunication systems will enable full remote monitoring and control from the Ekurhuleni Distribution Control Centre and Eskom National Control, in accordance with 07_Grid_Connection_Requirements.pdf and IEC 61850.9.1 SCADA Architecture

- Three-level hierarchy: Process level (Merging Units & IEDs), Bay level (BCUs), Station level (redundant station computers).
- Communication: Fibre-optic Ethernet ring (IEC 61850-8-1 MMS & GOOSE) with PRP redundancy.
- Protocols: IEC 61850 for intra-substation; DNP3 for interface to existing Eskom SCADA.
- Data points: All analogue values, status, alarms, sequence-of-events and energy metering will be time-stamped and transmitted.

9.2 Metering Philosophy

- Main and check energy meters (Class 0.2S accuracy) on each 132 kV incomer and each 33 kV feeder, connected to dedicated CT/VT cores.
- Power-quality monitoring per NRS 048-2 (voltage, harmonics, flicker) at the 33 kV bus.
- All metering data will be integrated into the SCADA database and available for billing and settlement.

9.3 Telecommunication

- Dedicated fibre-optic link (provided under separate contract) from substation to nearest Eskom fibre node.
- Redundant radio backup link for emergency voice and SCADA.

10. PHYSICAL LAYOUT & CIVIL INTERFACE PHILOSOPHY

The substation will be an Air-Insulated Switchgear (AIS) outdoor yard at 132 kV and indoor metal-clad switchgear at 33 kV, with a single-storey control building. Layout philosophy is driven by SANS 1019 minimum clearances, operational access requirements, future expansion space and EIA constraints (wetland buffer, dust suppression).

10.1 General Arrangement Principles

- 132 kV yard: Portal structure with minimum phase-to-phase clearance of 1.8 m and phase-to-earth 1.1 m. Equipment arranged in a linear “breaker-and-a-half ready” configuration to allow future conversion if required.
- 33 kV switchgear: Indoor, metal-clad, double-bus arrangement in a single row for short cable runs.
- Transformer placement: Oil containment bunds with fire walls and 10 m separation from adjacent plant.
- Spare bays: Fully civil-prepared (foundations, trenches, cable ducts) for future third transformer and four additional 33 kV feeders.

10.2 Civil Interface Requirements

- Cable trenches: Precast concrete with removable covers, sized for all current and future circuits plus 30 % spare capacity.
- Access roads: 6 m wide main roads and 4 m wide secondary roads with 20 t axle loading.
- Drainage: Graded yard with oil/water separator and connection to municipal stormwater system.
- Control building: Located outside the high-voltage yard, with separate HVAC, battery room and relay room. All civil designs will be based on the geotechnical report (180 kPa bearing capacity) and wind/seismic loading per SANS 10160.

10.3 Safety & Operability

Minimum 4 m clearance around all live equipment for maintenance vehicles. Lifting beams and hardstanding areas provided for transformer and breaker replacement without crane out-of-yard movement.

11. AUXILIARY POWER & DC SUPPLY PHILOSOPHY

11.1 AC Auxiliary Supply

Station AC supply will be derived from two independent 415 V sources:

- One 500 kVA auxiliary transformer tapped from each main power transformer tertiary (11 kV).
- Automatic change-over with < 100 ms interruption.
- Distribution via duplicated LV switchboards feeding lighting, heating, small power, HVAC and fire systems.

11.2 DC Supply System

- Nominal 110 V DC system with two independent battery banks (each sized for 8-hour backup at full load plus 30 % margin).
- Two 50 A float/boost chargers per bank with automatic change-over.
- DC distribution boards duplicated and segregated for protection, control and SCADA loads.
- Battery type: VRLA (valve-regulated lead-acid) in accordance with Eskom TRS requirements.

11.3 UPS System

Separate 230 V AC UPS (30 kVA, 30-minute backup) for critical SCADA, telecommunication and HMI loads. All auxiliary systems are designed so that the loss of any single AC or DC source does not affect protection or control functionality, maintaining full N-1 compliance at the auxiliary level.

12. CONSTRUCTABILITY, OPERABILITY, MAINTAINABILITY & SAFETY

The substation layout and detailed design have been developed from the outset with full consideration of constructability, operability, maintainability and safety in accordance with SANS 1019, Eskom TRS requirements, the Occupational Health and Safety Act (OHSA) and ORHVS (Operational Regulations for High Voltage Systems). All philosophies locked in earlier sections are translated into practical construction and operational requirements to ensure the project can be delivered safely, on schedule and within the ZAR 195 million budget.

12.1 Constructability

Construction sequencing will follow a modular bay-by-bay approach. The 132 kV yard will be built in two phases: first the Main Bus and incoming line bays, followed by the Transfer Bus and transformer bays. This allows early energisation of one transformer and partial 33 kV supply. Pre-cast cable trenches, foundations and gantry structures will be used wherever possible to reduce on-site concreting time. All major equipment (transformers, reactors) will be delivered by road with pre-arranged heavy-lift routes and temporary hardstanding areas. Construction access roads (6 m wide main, 4 m secondary) will be completed in the first mobilisation phase to support 20 t axle loads. Dust suppression, wetland buffer compliance and limited working hours (07:00–17:00 weekdays per 04_EIA_Report.pdf) are embedded in the construction method statements.

12.2 Operability Philosophy

The substation will be fully operable from the local control room or remotely via SCADA. All circuit breakers and motorised disconnectors will be fitted with local control panels and position indicators visible from ground level. Comprehensive mechanical and electrical interlocking (both hardwired and IEC 61850 GOOSE) will prevent unsafe switching sequences. Synchro-check supervision is mandatory before any breaker closure. Emergency trip push-buttons will be located at strategic yard points and inside the control building. The design allows any single circuit or transformer to be taken out of service without interrupting the remaining N-1 capacity.

12.3 Maintainability

Hot maintenance on busbars and couplers is facilitated by the selected Main & Transfer (132 kV) and Double Bus (33 kV) schemes. Each bay is provided with permanent hardstanding for mobile maintenance equipment and 4 m minimum clearance around live parts for safe working. Transformer and reactor oil sampling points, Buchholz relays and OLTC mechanisms are located at ground level. Spare bays are fully civil-prepared (foundations, trenches, earthing) so that future extensions can be executed live. A dedicated maintenance workshop and stores area is included in the control building footprint. All equipment is selected for a minimum 40-year design life with standard Eskom spares availability.

12.4 Safety

Personnel safety is the overriding priority. The earthing grid (Section 7) ensures step and touch voltages remain within IEEE 80 limits under maximum fault conditions. All live equipment is protected by surge arresters and adequate clearances per SANS 1019. Warning notices, anti-climb fencing and locked gates comply with ORHVS. An integrated fire detection and suppression system (CO₂ for control room, water mist for transformers) is provided. A full Hazard Identification and Risk Assessment (HIRA) will be completed during detailed design, and a Construction Safety File will be maintained throughout the project.

13. HIGH-LEVEL COST, SCHEDULE & OPTIMISATION

The design decisions in this DBR have been optimised to deliver the required N-1 performance and future-proofing within the approved ZAR 195 million (excl. VAT) budget envelope stated in 08_Project_Budget_&_Scope.pdf.

13.1 Capital Cost Breakdown (High-Level)

- Primary plant & switchgear (132 kV & 33 kV): 48 %
- Transformers, reactor and capacitor bank: 18 %
- Civil works, foundations, trenches and control building: 15 %
- Protection, control, SCADA and auxiliary systems: 10 %
- Earthing, lightning protection and cabling: 6 %
- Contingency & engineering: 3 %

13.2 Value Engineering & Optimisation

The selection of Main & Transfer Bus at 132 kV (instead of 1½ Breaker) and Double Bus Single Breaker at 33 kV saved approximately ZAR 22 million in switchgear and civil works compared with higher-cost schemes while still meeting full N-1 requirements. Use of indoor 33 kV switchgear reduced the yard footprint by 25 %, lowering land preparation and earthing costs. Standardisation on Eskom-preferred vendors (where pre-approved) further reduces spares holding and training costs for the Client. All future expansion provisions (third transformer, additional feeders) are civil-prepared only, deferring major expenditure until load growth materialises.

13.3 Project Schedule (High-Level)

- DBR approval: Month 1–2
- Detailed design & procurement: Months 3–9
- Site mobilisation & civil works: Months 10–14
- Equipment installation & testing: Months 15–20
- Commissioning & handover: Months 21–24
- Total duration: 24 months from DBR approval to commercial operation, allowing for long-lead transformer procurement (12 months).

The design remains within budget while delivering the exact performance required by 01_Network_Planning_Report.pdf.

14. RISKS, MITIGATIONS & ASSUMPTIONS

14.1 Identified Risks & Mitigations

Risk Description	Likelihood / Impact	Mitigation / Residual Risk
Higher-than-expected fault levels due to network changes	Medium / High	Confirm final fault levels in detailed short-circuit study; equipment ratings include 10% design margin
Soil resistivity variation across site	Medium / Medium	Additional Wenner tests during construction; contingency counterpoise included in BOQ
Delay in incoming 132 kV line completion	High / High	Staged energisation plan using one transformer initially; coordination with Eskom Lines contract
Environmental delays (wetland or heritage)	Low / High	EIA already approved; buffer zones and environmental monitoring plan in place
Long-lead equipment (transformers)	Medium / High	Early procurement strategy recommended; multiple OEMs pre-qualified
Budget overrun due to currency fluctuation or escalation	Medium / Medium	Escalation allowance (~8%) included; fixed-price tender strategy to limit exposure

14.2 Key Assumptions

- Load forecast in 01_Network_Planning_Report.pdf remains valid.
- Incoming 132 kV lines and fibre links will be available on the agreed programme.
- No major changes to Eskom TRS or Grid Code during detailed design.
- Geotechnical conditions are as reported (no unforeseen rock or high water table).
- Client will provide all necessary wayleaves and permits prior to mobilisation.

Any deviation from these assumptions will be formally notified to the Client with impact assessment.

15. CONCLUSION & PATH FORWARD

This Design Basis Report establishes a complete, auditable and optimised basis for the Klipfontein 132/33 kV Substation that fully satisfies the N-1 reliability, power quality, environmental and budget requirements defined in the Data Provided package. The selected Main & Transfer Bus (132 kV) and Double Bus Single Breaker (33 kV) configuration, combined with the bay allocation, reactive compensation and earthing philosophy, represents Eskom-standard practice for this voltage class while delivering operational flexibility, maintainability and future-proofing. All subsequent detailed design deliverables — including the full Single Line Diagram, General Arrangement layout, protection coordination study, earthing calculations, civil/structural drawings and BOQ — will be developed strictly in accordance with the philosophies and decisions locked in this DBR.

Recommended Path Forward

1. Client / Eskom review and approval of this DBR (target: 4 weeks).
2. Issuance of Notice to Proceed on detailed design.
3. Commencement of procurement for long-lead items (transformers and reactors).

The design team is confident that the Klipfontein 132/33 kV Substation will be delivered as a safe, reliable and cost-effective addition to the Gauteng East network.

Prepared by:

Hillary Mangena (Electrical)

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